

问题 A: 特技演员 影片在拍摄中, 一个激动人心的动作场景将要摄入镜头, 而你是特技协调员! 一位特技演员驾驶着摩托车跨越一头大象, 随后跌落在借以缓冲的一堆纸箱上. 你需要保护特技演员, 而且, 也要使用相对而言较少的纸箱 (较低的花费, 不能进入镜头, 等等)。 你的工作如下:

- ? 确定所用纸箱的大小
- ? 确定所用纸箱的数目
- ? 确定纸箱的堆放办法
- ? 还请确定, 通过对纸箱的各种调整, 是否会有所帮助
- ? 请把你的研究推广到不同组合重量 (特技演员 & 摩托车) 和不同跨越高度的情形 留心一下, 在影片 “明日帝国” 中, 角色 James Bond 驾驶着摩托车飞过一架直升机.

Problem:

An exciting action scene in a movie is going to be filmed, and you are the stunt coordinator! A stunt person on a motorcycle will jump over an elephant and land in a pile of cardboard boxes to cushion their fall. You need to protect the stunt person, and also use relatively few cardboard boxes (lower cost, not seen by camera, etc.).

Your job is to:

- determine what size boxes to use
- determine how many boxes to use
- determine how the boxes will be stacked
- determine if any modifications to the boxes would help
- generalize to different combined weights (stunt person & motorcycle) and different jump heights

Note that, in "Tomorrow Never Dies", the James Bond character on a motorcycle jumps over a helicopter.